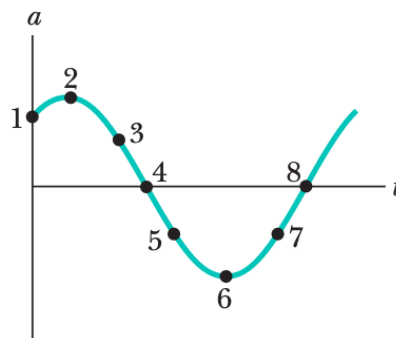


PYL1002: Waves and Oscillations – Mid Semester Exam

1. This exam is timed and will last **2 hours** from 3:30 - 5:30 pm. Total marks are 40.
2. Strict adherence to the **Honor Code** policy of the class is expected. Among other things, you are not allowed to collaborate with others in any form or fashion.
3. There are a total of 4 questions (each with multiple parts). Not all questions are of equal length, so please go through them all before you start and allocate your time appropriately. In particular, problem 1 is a collection of 4 independent, quick conceptual questions which should not require too much math. Some problems may seem lengthy but often the text provided is there to help you.
4. Please justify your answers and show all work.

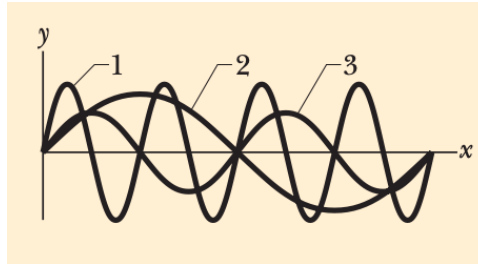
1 Some Quick Conceptual Questions [10 points]

- a [2 points] Which of the following relationships between the force F on a particle of mass $m = 0.1\text{kg}$ and the particle's position x gives simple harmonic motion (SHM): (i.) $F = -10x$, (ii.) $F = 150x^2$, (iii.) $F = 10 \cos(5x)$? For the case with SHM, what is the time period T of oscillation? Justify your answer. You can take $\pi \approx 3$ if needed.
- b [3 points] The following graph shows the acceleration $a(t)$ of a particle undergoing simple harmonic motion. (i.) Which of the labeled points corresponds to the particle at $x = -A$ where A is the amplitude of oscillation? (ii.) At point 8, is the velocity of the particle positive, negative, or zero? (iii.) At point 3, is the particle at $x = -A$ or $x = +A$, or between $-A$ and 0 , or between 0 and $+A$? Justify your answers.



- c [2 points] Two identical pendulums are coupled by a spring (just like we did in class). One normal mode has both masses moving in phase and the other has them moving out of phase. Without calculation, which mode has the higher frequency? Explain and justify physically.

- d [3 points] The figure is a composite of three snapshots, each of a wave traveling along a particular string. The phases (the argument which enters the sin or cos of a wave) for the waves are given by (a.) $2x - 4t$, (b.) $4x - 8t$, and (c.) $8x - 16t$. (i.) Which phase corresponds to which wave in the figure? (ii.) Rank the three waves according to their speed, greatest first. Justify your answers.



2 The Power Line [10 points]

When you suspend a heavy chain from fixed supports at both ends, the resultant curve is called a “catenary.” The equation of a catenary is given by

$$y(x) = a \left(\cosh \left(\frac{x}{a} \right) - 1 \right), \quad (1)$$

where $a > 0$ is a positive constant, x is the horizontal coordinate and y is the vertical coordinate. You would have seen this shape more often than you think. Think about every time you see a massive chain suspended on two end points. In particular, freely-hanging overhead power lines also form a catenary – most prominently visible with high-voltage transmission lines.



In case you want a refresher in hyperbolic trigonometry, here is how cosh is defined,

$$\cosh(x) = \frac{e^x + e^{-x}}{2}. \quad (2)$$

Now, suppose a mass m slides *without friction* along the catenary arc of a power line under the influence of gravity.

- a [2 points] What is the gravitational potential energy $U(x, y)$ of the particle on the catenary curve, assuming the potential energy at the origin $(0, 0)$ is zero? Using the catenary equation $y = y(x)$, write the potential energy only as a function of x , *i.e.* $U = U(x)$.
- b [2 points] At what point on the catenary does the net force on the particle vanish? What is the value of the normal force on the particle at this point?

- c [4 points] Expand the potential energy $U(x)$ from part (a) around this point of local equilibrium to show that for small deviations around equilibrium, the particle will execute simple harmonic oscillation. Make sure you show all steps and details clearly.
- d [2 points] What is the angular frequency of oscillation ω of this particle undergoing small oscillation?

3 The Power Line, Again – With Friction [10 points]

Note: This problem builds off the American Journal of Physics article on constant-magnitude friction force damping. If you use a result from the paper, quote and cite it clearly.

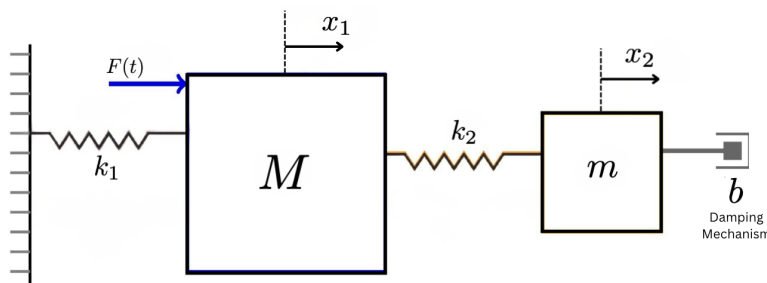
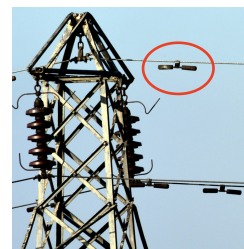
We continue our discussion from Problem 2 above. Most power lines are made of rough material and hence have *friction* on them. The coefficient of static friction is μ_s and that of kinetic friction is μ_k with $\mu_s \geq \mu_k$ as usual. We wish to gain an understanding of how the particle would slide on this rough power line under friction. You can ignore viscous damping (the one proportional to particle speed which we have discussed in class).

- a [3 points] For small oscillations, the motion may be reduced to an effective 1D problem along the x -direction, as in Problem 2. Complete this argument by briefly showing the following:
 - (i.) the friction force, while tangent to the catenary curve, can effectively be taken to point along the x -direction to leading order.
 - (ii.) Show that the normal force on the particle to leading order is $N \approx mg$ at any point for small oscillations.
 - (iii.) Show that the effective “spring constant” k for the resulting oscillations is $k = mg/a$.
- b [2 points] Write down the governing equation of motion ($F = ma$) for this particle. You may need to write two equations depending on the sign of the particle’s velocity. You do not need to solve them.
- c [1 point] What is the minimum displacement $x_0^{(\min)}$ for the particle along x -direction for it to start moving if it begins at rest?
- d [2 points] Say you start the particle at rest from an initial displacement $x_0 > x_0^{(\min)}$ at $t = 0$. What is the condition on x_0 for the particle to cross the origin at least once? In terms of other variables given in the problem, what is the time taken for the particle to stop for the first time after it begins motion?
- e [2 points] For a typical long distance power-line, $a = 500$ m, and the coefficients of friction are $\mu_s = 0.5$ and $\mu_k = 0.4$. If you start the particle of mass $m = 1$ kg from $x_0 = 450$ m from the origin, how many times does it cross the origin? Find the time it takes for the particle to come to a complete stop. You can use $g \approx 10$ m/s², $\pi \approx 3$, and $\sqrt{50} \approx 7$, if needed.

4 The Power Line, Yet Again – With a Tuned Mass Damper [10 points]

Let us continue our saga with the Power Line. **Now, we will focus on the dynamics of the power line as a whole, not a particle moving on it.** Power line cables are suspended in the air for long distances and are very vulnerable to periodic swaying and flutter induced by the wind. If the natural frequency of oscillation of the power line matches with the periodic driving frequency of the wind, resonance takes place and the power line can undergo catastrophic swaying movement.

To counter this, a *tuned mass damper* is attached on the power line. Here is an example of the so-called “Stock bridge Damper.” The basic idea is to attach a secondary oscillator whose natural frequency is matched to the primary system, making it resonate out of phase and siphon away energy, thereby reducing the main system’s vibration amplitude.



The figure above is the schematic model we will work with. The power line is modeled as the primary system of mass M connected to spring k_1 . It is driven by the wind by a periodic driving force $F(t) = F_0 \cos(\omega t)$. The primary mass is connected to a secondary mass m via the spring k_2 , and this secondary mass is damped by viscous damping $F_{\text{damp}} = -b\dot{x}_2$. There is no damping on the primary mass M . You do not need to account to gravity separately.

- a [3 points] Write down the equations of motion ($F = ma$) for both the primary mass M whose displacement from equilibrium is given by $x_1(t)$, and the secondary mass m which has a displacement $x_2(t)$ from its equilibrium. Work in the complex plane, so that $F(t) = F_0 e^{i\omega t}$.
- b [2 points] Look at the *steady state* behaviour for both the masses when they respond at the wind’s driving frequency, that is, $x_1 = A_1 e^{i\omega t}$ and $x_2 = A_2 e^{i\omega t}$. Here, A_1 and A_2 are complex amplitudes¹. Write the solution in the following form, and determine a, b, c, d, e, f .

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix}$$

The natural frequency of oscillation of the primary mass alone is $\omega_0 = \sqrt{k_1/M}$. In the absence of the secondary mass, resonance for mass M would have occurred when the wind frequency matches the natural frequency, $\omega \approx \omega_0$. We want to understand how the addition of the secondary mass takes energy away from the primary. In the following parts, you may take $\omega = \omega_0$:

- c [2 points] For simplicity, assume the damping coefficient b is very small ($b \approx 0$ for this part), and the wind frequency matches the primary’s natural frequency, $\omega \approx \omega_0$. What value of k_2 (in terms of m, M and k_1) will give $A_1 = 0$? This is the *tuning condition* where the primary mass does not move.
- d [2 points] Now taking into account a small but nonzero damping coefficient b , and show that, at the tuning condition from part (c), we get for the amplitude ratio to be $A_1/A_2 = ib/(m\omega_0)$.
- e [1 point] Use the result from part (d) to argue that the oscillation of the two masses is out of phase by an angle of $\pi/2$. Remember A_1 and A_2 are complex numbers.

¹For example, $A_1 = C_1 e^{i\phi_1}$ for some real amplitude C_1 and real phase ϕ_1 , and similar for A_2